

Drone (vs) UAV :-

The term Drone & UAV are often used interchangeably, but they have slight different meaning.



UAV

An Aircraft that

- flies without a human pilot onboard.
- is controlled remotely or operated autonomously.

Uses:-

- 1) Aviation regulations
- 2) Engineering tools
- 3) Military & research work

Drone

- A general term used for
- Unmanned flying devices [Ground + underwater robots]
- Especially small quadcopters

Uses:-

- 1) daily conversation
- 2) media
- 3) Hobby & commercial use.

Aspect	UAV	Drone
Nature of term	Tech & official	Informal & general
Usage	Engineering, Aviation, Military	Common / public use
Scope	only <u>Aerial vehicle</u>	Can include <u>aerial</u> , <u>ground</u> , <u>underwater</u> robots (in casual use)

Type	Mechanics	Strength	Idealness	Common-Real world ways
Multi-Rotor	• Multiple propellers • V-TOL	• Easiest to fly, stable, precise control in tight spots	• Short battery life, • Lower speed • Limited range	→ Media & Security - film, weddings, - real estate - tower inspections
Fixed wing	• Airplan design • Uses wings for lift via forward motion	• Long Endurance, • high speed, • cover massive area	• Requires a Runway • cannot hover	→ Large-scale task • Agriculture • Land surveying • border patrol
Hybrid (VTOL)	• Merges rotors for takeoff with wings for forward flight	• No runway needed • Long range + flexibility • All-terrain	• Expensive • Complex design • Harder to repair	→ Specialized logistics • Search & rescue • military • Remote delivery

ESC [Electronic Speed controller]: -

① ESC = Motor Speed controller

- Takes the power from the battery [DC-power]
- Uses the signals from the flight controller
- Controls how fast the motor spins

② ESC Rapidly switches power [PWM]

ESC don't just smoothly vary voltage, instead,

- ESC turns power on & off very fast, is called as PWM [Pulse width Modulation].

More on time = faster the motor speed

More off time = slower the motor speed

This switching b/w on & off is the heart of the ESC behavior

③ It creates -3 phase power for brushless motor.

Drone motors are brushless, so, they need:

- Three alternating electrical phases.

The ESC, Sends current in a precise sequence, this creates a rotating magnetic field that spins the motor.

No ESC = motor won't spin at all

ESC don't just set speed once, they thousands of times per second, Respond to flight controller commands

GPS [Global positioning System]:-

↳ Satellite-based Navigation System that enables the drone to determine its exact location on Earth.

→ A GPS receiver in the drone collects the GPS signals from the multiple satellites orbiting ~~area~~ the earth. & it calculates its position using a method called "trilateration". the drone's flight controller uses this data to track the location & movement.

GPS provides:-

Latitude → N-S position

Longitude → E-W position

Altitude → Height above the ground/sea level.

(*) Role of GPS in advanced drone functions:-

1. Position Hold [Hovering]

2. Navigation → Helps the drone move to a specific location

→ Enables waypoint-based flying
[following a set path]

3. Return to Home [RTH].

↳ in case of Signal lost / low battery.

↳ Measure Speed & Distance tracking

(*) Limitations of GPS:-

↳ Signal may be weak near tall buildings or indoors. ↳ Accuracy can be affected by weather & obstacles. ↳ takes few sec to lock onto satellites.
~~conclusion~~

IMU [Accelerometer + Gyroscope]:-

↳ A Sensor used in drones to understand their movement & orientation.

Accelerometer:- Measures linear acceleration, which means how fast the drone is slowing down & speeding up

in

X-axis → forward & backward

Y-axis → left & right

Z-axis → up & down

→ It also detects the "Gravity" which helps determine the drone tilt.

Gyroscope:- Measures angular velocity, which means how fast the drone is rotating.

Rotation around three axis:-

(i) Roll → tilting left or right.

(ii) Pitch → tilting forward or backward.

(iii) Yaw → turning left or right.

→ Accelerometer

- Helps if drone is tilting
- Assists in maintaining balance
- Helps in smooth movement and hovering.

→ Gyroscope

- Track rotation & Orientation
- Helps keeps the drone steady
- Enables smooth turns & precise control.

PID [Proportional - Integral - Derivative] :-

↳ A core control algorithm that helps a drone stay stable and respond smoothly to commands.

→ PID loop runs hundreds (or thousands) of times per sec inside the flight controller. It constantly adjusts motor speed to control

→ Pitch (forward / backward tilt)

→ Roll (left / right tilt)

→ Yaw (left / right turn)

1. Proportional (P) :-

* Reacts to current error

* If drone tilts, "P" pushes it back proportionally

* Too high → shaky / oscillations

* Too low → sluggish response.

2. Integral (I) :-

* Fixes accumulated past errors.

* Helps with drifts (like slow leaning due to wind)

* Too high → slow wobble @ overshoot

* Too low → drift remain

3. Derivative (D) :-

* Predicts future error (based on how fast things are changing)

* Adds damping → smooths motion

* Too high → jittery / noisy

* Too low → overshoot & bounce.

MAVLink [Micro Air Vehicle Link]:-

↳ Lightweight communication protocol used in drones to let different components talk to each other.

Components like

1. Flight controller (autopilot)
2. Ground Control station (laptop/phone)
3. Onboard devices (GPS, sensor, camera)

Features:-

- (*) Lightweight (works on low bandwidth)
- (*) Fast communication
- (*) Reliable (user error checking)
- (*) Open Source

How does it work:-

↳ Data sent as messages (packets)

↳ Each message contains:

1. Message ID
2. ~~Data~~ Payload (Data)
3. Checksum (for error detection)

Flight controller:- A compact onboard computer. ①

↳ It processes sensor data and user commands to keep the drone stable & controllable in the air.

Main functions:-

1. Stabilization:

- (*) keeps the drone balanced in all directions (P, P, P)
- (*) Automatically corrects tilt using sensor data

2. Control Processing:

- (*) Converts pilot commands into motor actions.

3. Sensor Integration:

- (*) Reads data from sensors like gyroscope, GPS etc

4. Motor Control:

- (*) Adjust speed of ^{each} motor to maintain flight stability & direction.

5. Autonomous Flight:

- (*) Supports GPS-based navigation, waypoint flying, return to home (RTH).

Key Components of Flight Controller:-

1. Microcontroller (MCU)

↳ Microprocessor runs flight software like Betaflight, ArduPilot, INAV, PX4

2. Sensors:

(2)

- (*) Gyroscope
- (*) Accelerometer
- (*) Magnetometer
- (*) Barometer
- (*) GPS module

3. ESC connections:

↳ Send signals to ESC to control motors.

4. Communication Ports:-

- ↳ Receiver i/p (radio signals)
- ↳ USB for configuration
- ↳ UART/I2C/SPI for sensor & modules.

Common flight controller Software:- (Autopilot-firmware)

- 1. Betaflight → Racing drone
- 2. Ardupilot → autonomous drones, research.
- 3. INAV → GPS navigation drones
- 4. PX4 → professional / Autonomous system

Autopilot firmware:-

↳ "A Software running on a drone flight controller" that enables:

- *) Automatic flight control
- *) Stability & Navigation
- *) GPS-based mission
- *) Sensor fusion & decision making.

↓
Pixhawk, Cube-etc.

It turns the basic flight controller into a "Smart autonomous system."

Popular Autopilot Firmware:-

1. ArduPilot :

- | Features | Supports |
|-----------------------------|---------------------------------|
| → Advanced mission planning | → Drones |
| → Terrain following | [Multi rotor, Fixed wing, VTOL] |
| → Precision landing | → Boats & rovers |
| → Fail safe system | |

(*) used in Autonomous drones & research.

2. PX4 :

- | Features | |
|--------------------------------------|---|
| → Real time flight control | → Modern & Modular autopilot System |
| → Smooth sensor fusion | → Used in professional drones & robotics |
| → High performance navigation | → Strong integration with simulation tools [ROS, Gazebo]. |
| → Supports commercial UAV platforms. | |

IMU [Accelerometer + Gyroscope]:-

~~IMU~~ PS

3. iNav [iNAV]:-

- ↳ Used mainly for GPS-assisted FPV drones
- ↳ Easier than PX4, Ardupilot

Features

- RTT+ (Return To Home)
- Position Hold
- Waypoint navigation (basic)

4. Betaflight :- [Not full autopilot]

- Used in manual (racing) drones.
- Limited autonomy (not full GPS mission).

Ground Control software:-

used to configure & control autopilot firmware.

- Mission Planner (Ardupilot)
- QGroundControl (PX4 & Ardupilot)